

3(a)	$\rho = m / V$	C1
	$V = \pi d^2 L / 4$ or $\pi r^2 L$	C1
	weight = $2.7 \times 10^3 \times \pi (1.2 \times 10^{-2})^2 \times 5.0 \times 10^{-2} \times 9.81 = 0.60 \text{ N}$	A1
3(b)(i)	the point from where (all) the weight (of a body) seems to act	B1
3(b)(ii)	$W \times 12$	C1
	$(0.25 \times 8) + (0.6 \times 38)$	C1
	$W = (2 + 22.8) / 12$ $= 2.1 \text{ (2.07) N}$	A1
3(c)(i)	pressure changes with depth (in water) or pressure on bottom (of cylinder) different from pressure on top	B1
	pressure on bottom of cylinder <u>greater than</u> pressure on top or force (up) on bottom of cylinder <u>greater than</u> force (down) on top	B1
3(c)(ii)	anticlockwise moment reduced and reducing the weight of X reduces clockwise moment or anticlockwise moment reduced so clockwise moment now greater than (total) anticlockwise moment	B1